

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 4

Algorithms and flowcharts

29 questions from 25 question papers, Feb/March 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

- Write an algorithm in pseudocode to enter and find the average of a set of numbers using a REPEAT...UNTIL loop. Your algorithm must work for different sets of numbers.

[8]

6 Many houses in cooler countries have central heating systems. In a central heating system, microprocessors are used to control the pump which sends water from the boiler to the individual heaters.

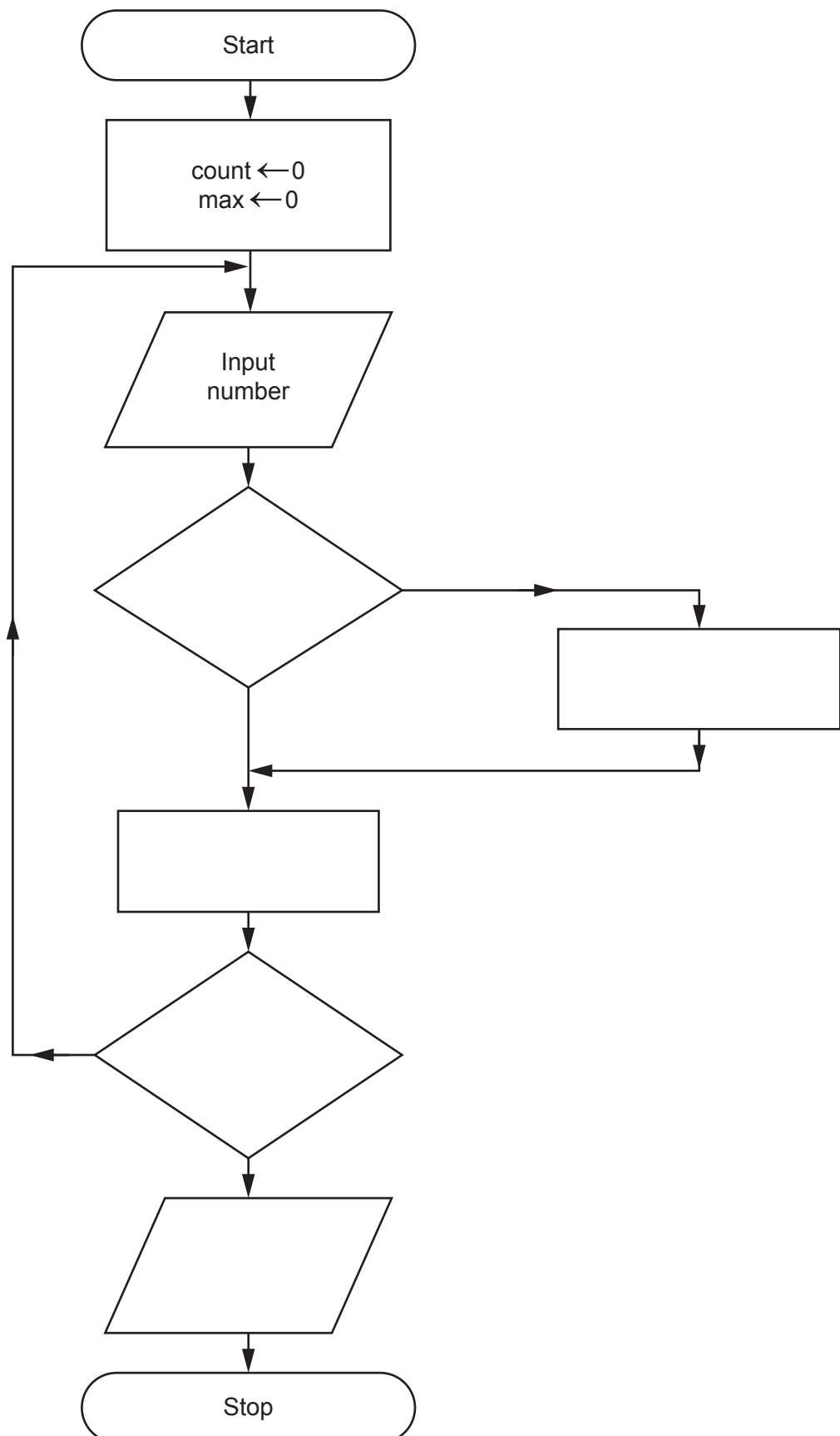
Complete the pseudocode algorithm to show the processing which takes place in this microprocessor-controlled central heating system. The algorithm must prevent attempting to switch the pump on when it is already on and attempting to switch the pump off when it is already off. You may assume `temperature` is the variable representing the actual temperature of the house and `preset` represents the required temperature of the house.

```
REPEAT
INPUT temperature
    IF temperature < preset
        THEN
```

[6]

- 7 Josefine has started to draw a flowchart which inputs 10 numbers and outputs the largest value.

Complete the flowchart by filling in the empty boxes and by placing yes and no as appropriate in the diagram.



[7]

Every week the master file has to be updated to include the new wage earned so far this year.

```

READ first record in transaction file
READ first record in old master file
WHILE not end of transaction file
    IF transaction file WorkerID = master file WorkerID
        THEN

```

..... [6

(b) Updating a master file in a payroll system is an example of batch processing.

Discuss the advantages and disadvantages of using batch processing, compared to other types of processing, for a payroll system.

[8]

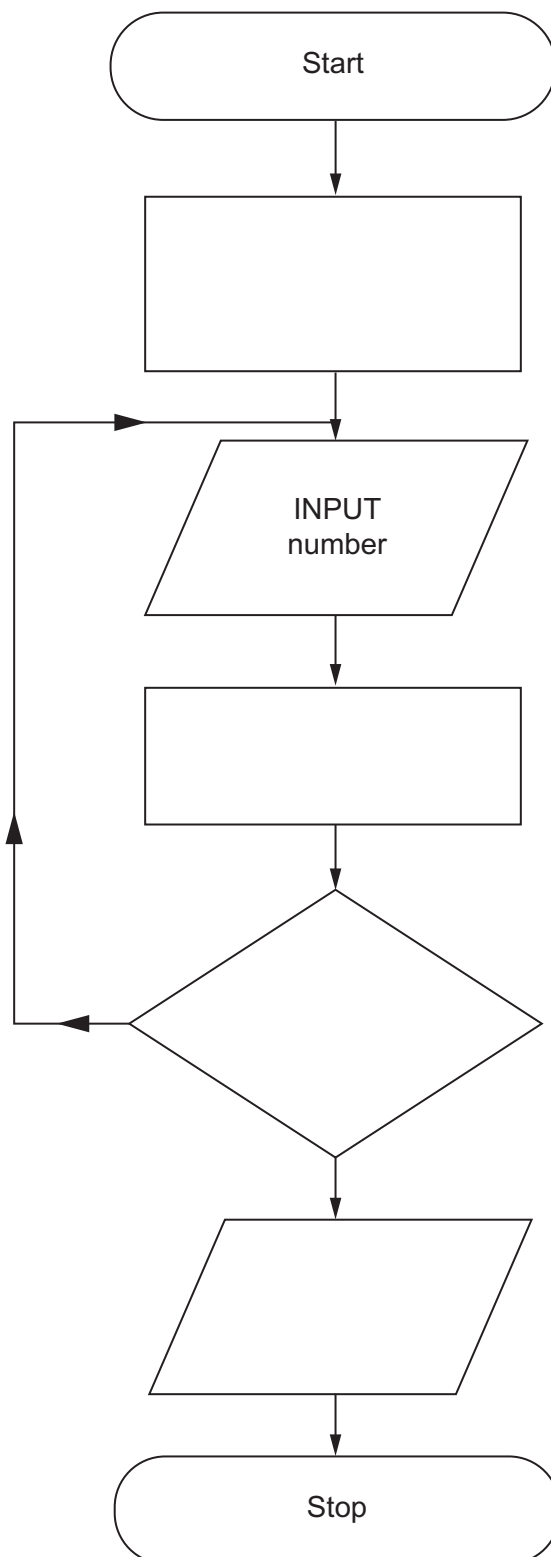
8 A microprocessor-controlled car park barrier is used to allow cars to enter a car park. When a car approaches the barrier, the barrier automatically rises to allow the car to enter. It does not lower until the car is safely clear of the barrier system. There is a light sensor and light beam source immediately before the barrier. No tickets are issued by the system. The system operates continuously while it is switched on.

Complete this pseudocode algorithm to describe the processing involved. Your pseudocode algorithm should include IF...THEN statements.

```
REPEAT
    INPUT reading from induction loop
```

..... [7]

- 11** Complete the flowchart below to add up 6 numbers.
Use the variable *total* to store the sum of the numbers and the variable *count* to control the number of times the loop is repeated.



[7]

- 9 Jose has a greenhouse to grow his plants. He lives in a country which has a warm climate so the greenhouse does not need a heater but has microprocessor controlled windows and a temperature sensor.

- (a) Describe how the sensor represents the temperature so the computer can read the data.

..... [1]

- (b)** Complete this pseudocode algorithm to show the processing which takes place to control the temperature of the greenhouse.

```
WHILE system switched on
  INPUT temperature
  IF temperature > preset
    THEN
      IF window closed
        THEN
```

[8]

- (c)** Discuss the advantages and disadvantages, to Jose, of using microprocessor control in this scenario.

[8]

- 9 A programmer is writing an algorithm in pseudocode to represent a company's payroll system. The system calculates a worker's wages before tax by multiplying the number of hours worked by the rate of pay per hour. A procedure within the main algorithm representing this could be:

```
PROCEDURE BeforeTax(Hours, Rate)
    WagesBeforeTax ← Hours * Rate
ENDPROCEDURE
```

There are two stages involved in calculating the wages after tax:

- the amount of tax paid by the worker is calculated by multiplying the WagesBeforeTax by the rate of tax (35%);
- the amount of tax paid is then subtracted from the WagesBeforeTax.

- (a) Write a procedure for calculating the wages after tax, assuming the value of WagesBeforeTax is passed to it.

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..... [4]

- (b) The programmer's algorithm will use the `BeforeTax()` procedure and the procedure created in part (a) to calculate and output the wage after tax for each worker.

Complete the algorithm. The statements have been numbered to help you.

```
1  count ← 0
2  INPUT NumberOfWorkers
3  REPEAT
4  .....
5  .....
6  .....
7  .....
8  .....
9  UNTIL ..... [6]
```

- 8 A teacher wishes to produce a computer program to output the grades awarded for **all** of her students.

If a student scores:

- more than 60 marks, they are awarded a grade A
- 50–60 marks, they are awarded a grade B
- 40–49 marks, they are awarded a grade C
- below 40 marks, they are awarded a grade D.

The teacher has written the following algorithm before writing the program. Unfortunately, there are errors and some lines have been left out (omitted). She has, however, managed to include the correct number of ELSE statements.

```

1.      INPUT mark
2.      IF mark > 60
3.          THEN
4.              PRINT "A"
5.          ELSE
6.              IF mark < 50
7.                  THEN
8.                      PRINT B
9.                  ELSE
10.                     IF mark >= 40
11.                         THEN
12.                             PRINT "C"
13.                         ELSE
14.                             PRINT "D"
15.                     ENDIF
16.                 ENDIF

```

Identify each error or omission and how these could be corrected. Line numbers have been included to help you.

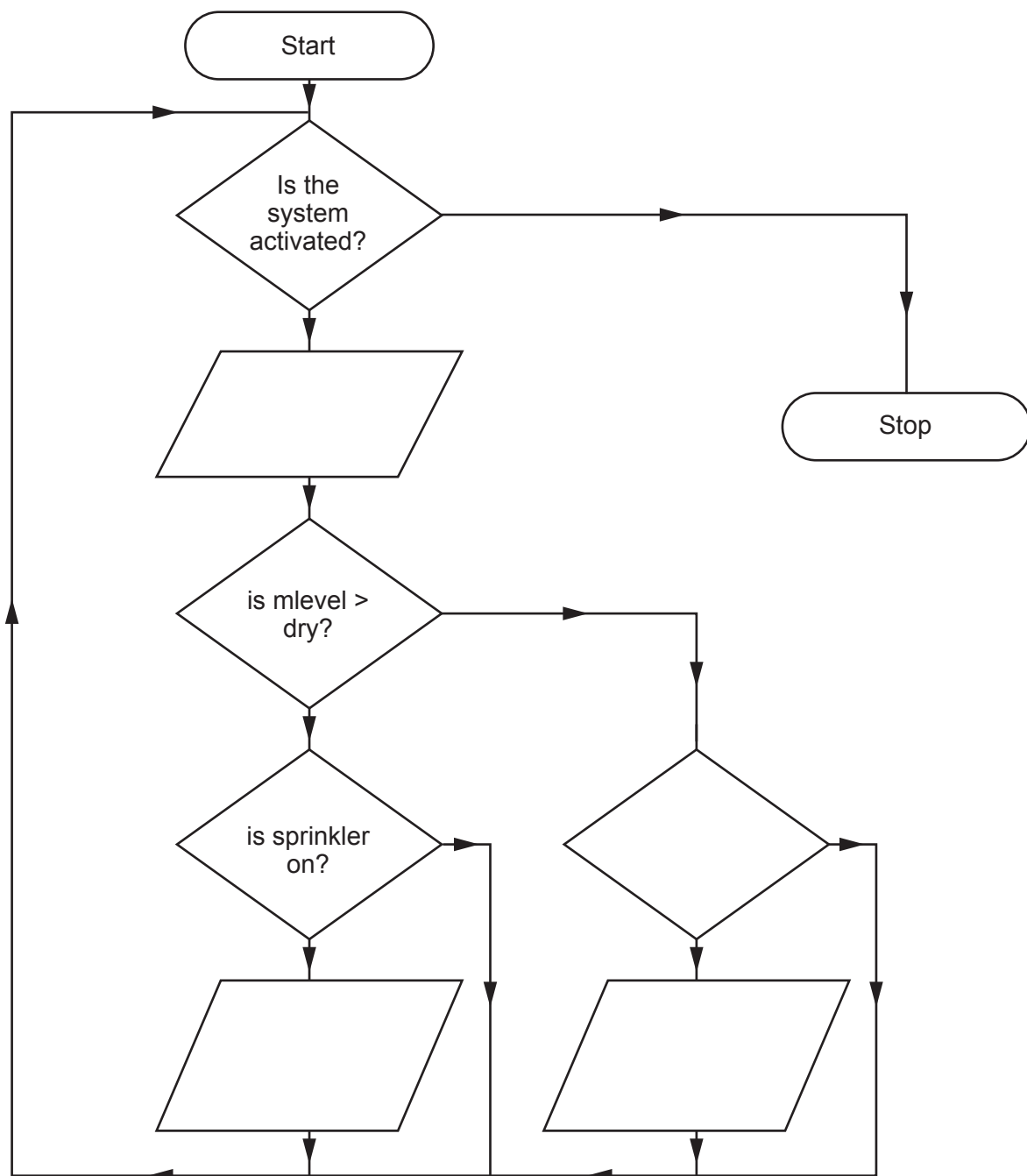
[6]

- 4 Mark has a small glasshouse. He lives in a warm country where plants can die if they are left without water for too long. The glasshouse has a microprocessor controlled sprinkler system and a moisture sensor in the soil.

Complete this flowchart to show the processing required to control the level of moisture in the soil of the glasshouse. You may assume that a pre-set value, **dry**, has already been input so that the system can test if the amount of moisture in the soil has reached or fallen below a certain level.

The user activates the microprocessor system by pressing a switch.

- (a) Complete the flowchart which inputs the moisture reading in the glasshouse and switches the sprinkler on and off according to this value. It also checks if the sprinkler is already on or off before attempting to switch it on or off.



[6]

(b) Flowcharts often contain subroutines.

(i) Draw the symbol used to represent a subroutine in a flowchart.

[1]

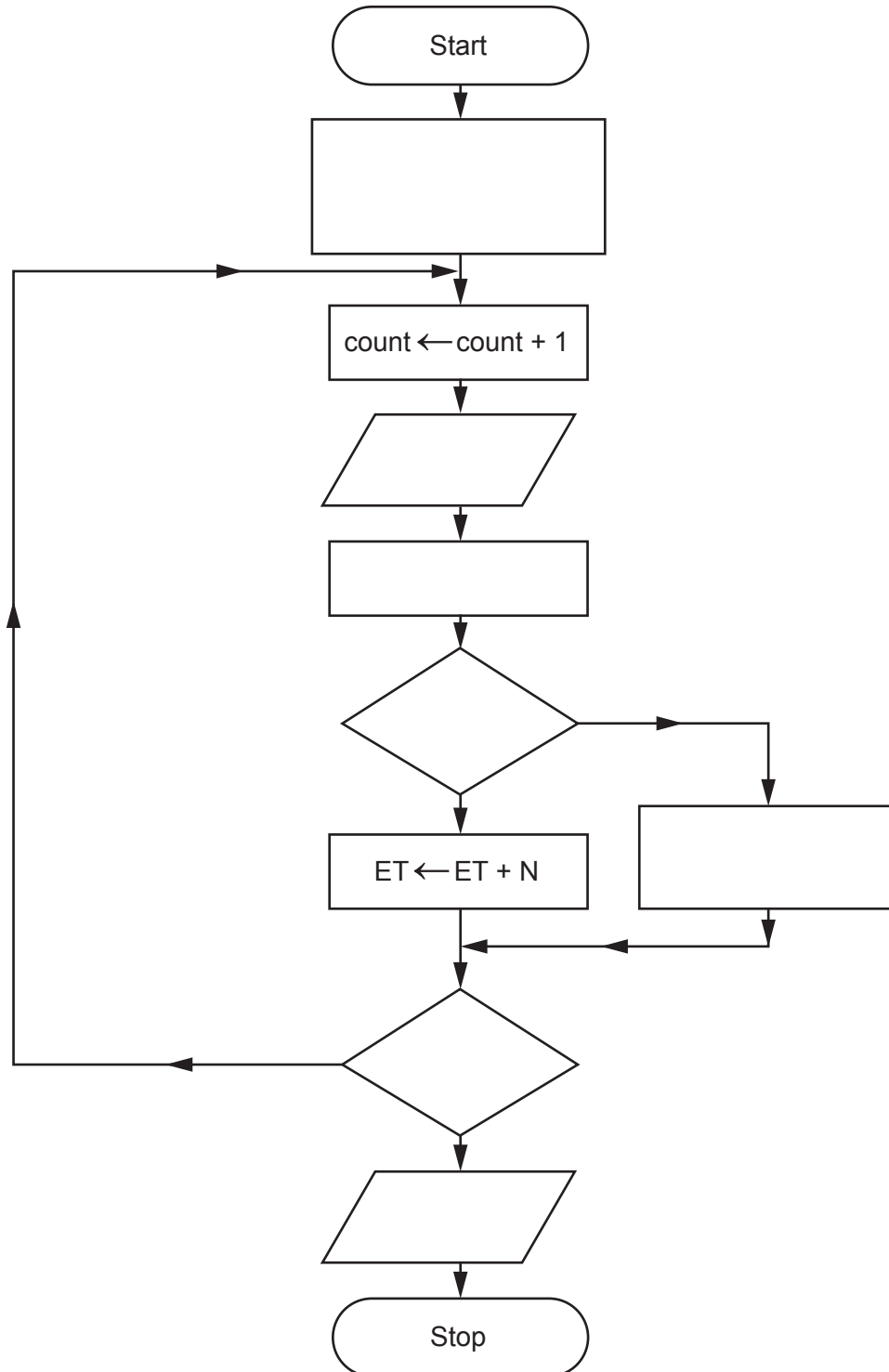
(ii) State what you would expect to see in the first box of a subroutine.

.....

..... [1]

8 Complete this flowchart which will allow the input of ten numbers.

- A number is even if the number divided by 2 is an integer.
- If a number (N) is odd, it will be added to the odd total (OT).
- If a number (N) is even, it will be added to the even total (ET).
- Both these totals will be output after the tenth number has been processed.



[8]

9626/12 May/June 2023 · Question 10

[Mark scheme](#)

- 10** Gustav has started to write an algorithm using pseudocode. It will input 6 numbers, all less than 1000, and output the smallest value.

Complete the algorithm using pseudocode.

```
count ← 0
smallest ← 1000
WHILE count < 6
```

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..... [6]

9626/13 May/June 2023 · Question 11

[Mark scheme](#)

- 11** In Ruritania these are the tax rules.

- People who earn up to \$10 000 pay no tax.
- People who earn between \$10 000 and \$40 000 pay 20% tax on their earnings **above** \$10 000.
- People who earn more than \$40 000:
 - pay \$6 000 ($\$30\,000 \times 20\%$)
 - and pay 40% tax on their earnings **above** \$40 000.

Complete this flowchart which shows how the net wage (the money earned after tax is paid) is calculated and is output for an individual in Ruritania.



- 9 Sukhjinder, a mathematics student, has defined a procedure using pseudocode. It calculates the area of a parallelogram by multiplying the base by the height. All parallelograms will have an area of less than 1000.

The procedure Sukhjinder has written is:

```
PROCEDURE parallelogram(base,height)
area ← base*height
ENDPROCEDURE
```

He has also begun writing an algorithm which calls the procedure and then passes back the area.

The main algorithm accepts inputs of the base and the height of 10 parallelograms, one at a time. It will output the areas of the largest and smallest parallelograms.

Complete the algorithm for Sukhjinder by filling in the missing lines.

```
count ← 0
smallest ← 1000
```

```
.....
WHILE count < 10
    INPUT base, height
```

```
.....
    IF area < smallest
        THEN
```

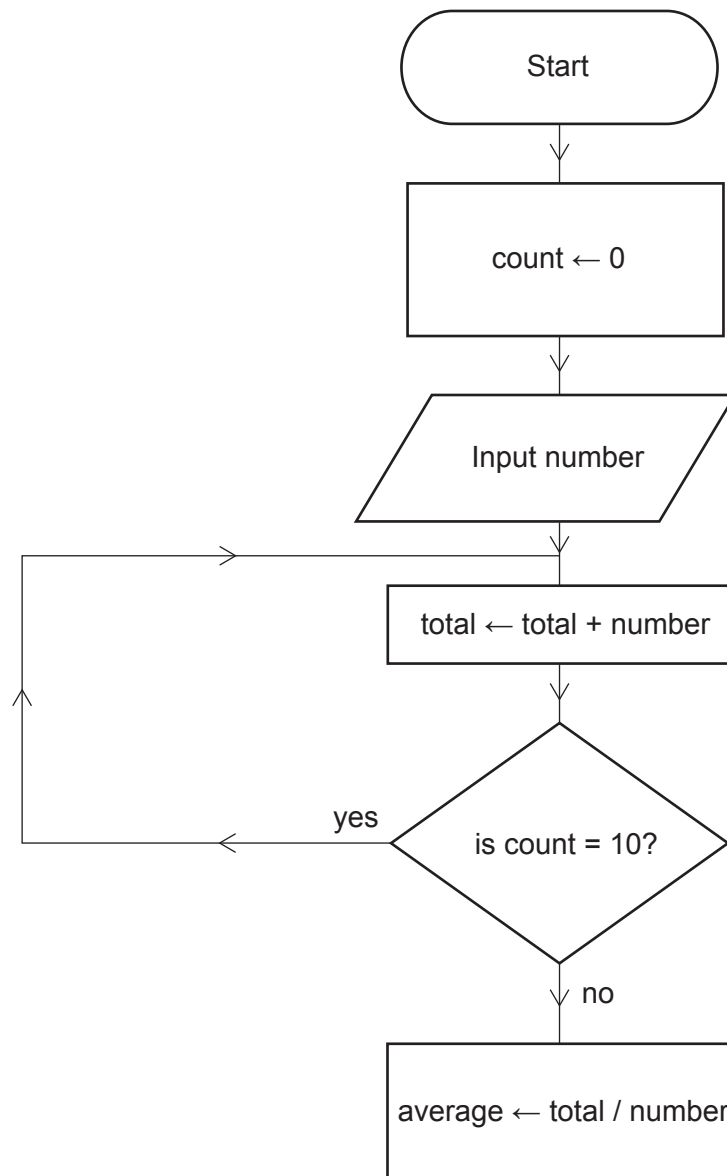
```
.....
        ENDIF
```

```
..... [8]
```

8 The average (mean) of a set of 10 numbers is calculated by:

- adding the numbers together to get a total
- dividing the total by 10.

Juan has attempted to draw a flowchart of an algorithm which finds the average of 10 numbers. He has made some mistakes and omitted to include some boxes.



Suggest **six** improvements he should make to his flowchart.

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..... [6]

- 9 A programmer is writing an algorithm in pseudocode to determine whether a number is a factor of another number.

Y is a factor of X if the result of X/Y is a whole number.

In pseudocode `INT (W)` returns the whole number part of W.

If W is the same as `INT (W)` that means W is a whole number.

Complete the algorithm, including the missing first line, in pseudocode. Each `THEN` or `ELSE` statement should be on a separate line to the action resulting from it. The algorithm uses:

- W to store the result of X divided by Y
- Z to store the value of `INT (W)`.

The algorithm must output the values of Y and X together with an appropriate message if Y is a factor of X or if Y is **not** a factor of X.

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`W ← X / Y`

`Z = INT (W)`

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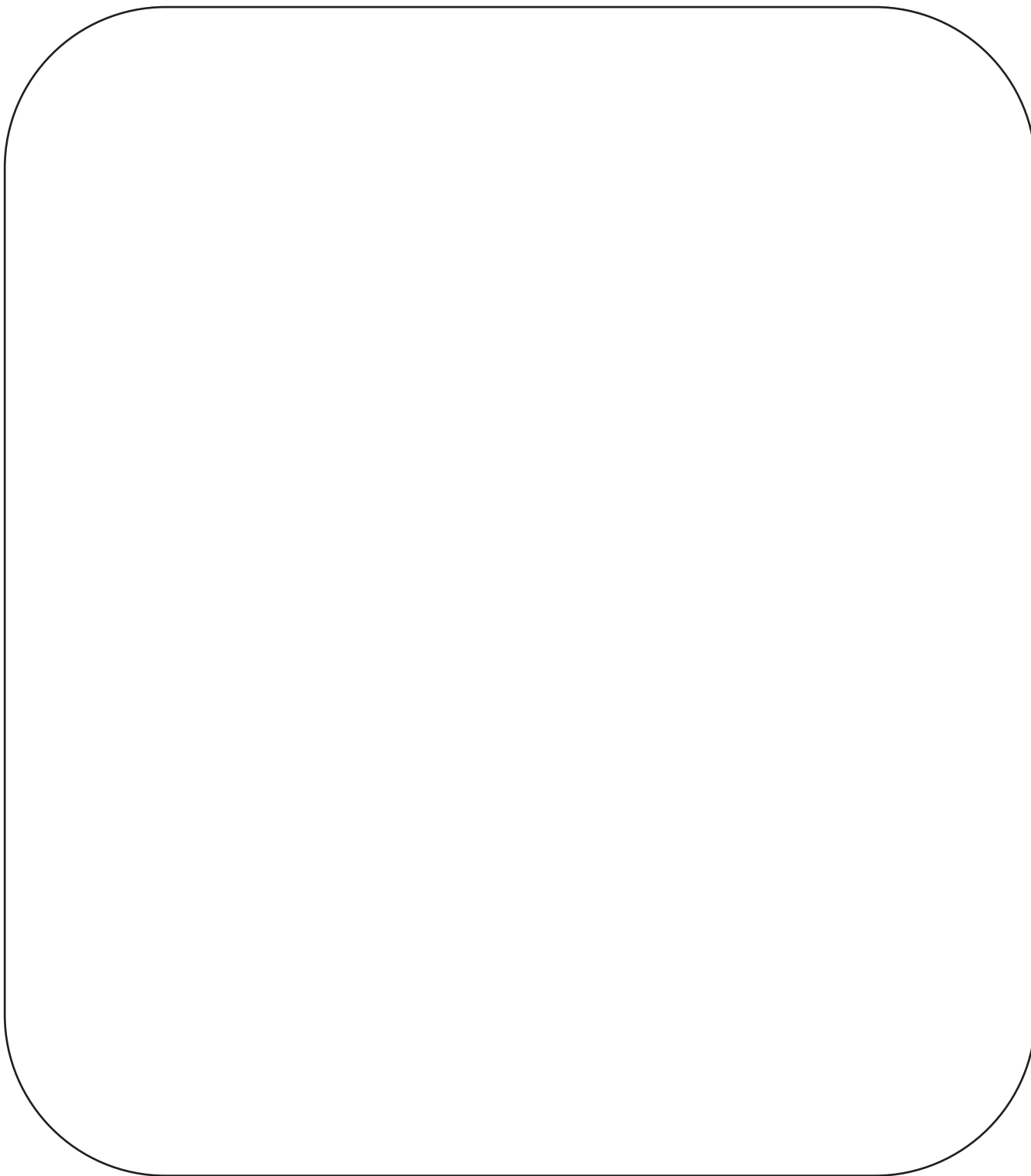
..... [8]

- 7 Draw a program flowchart, in the space provided, that will allow an examination mark to be entered and then output an appropriate message giving the correct grade, as shown in the table. All marks are integers.

Mark	Message
85 or more	Grade A
70 or more	Grade B
50 or more	Grade C
Any other mark	Grade U

4 Draw a program flowchart to represent this algorithm:

- Input an integer value.
- Repeat this until you have entered 20 values.
- At the end, output the largest value and the average of the values.



[7]

1 Different symbols are used in program flowcharts.

Draw, in the spaces provided, **four** different program flowchart symbols and for each one, state its purpose.

Symbol 1

Purpose

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Symbol 2

Purpose

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Symbol 3

Purpose

.....

.....

Symbol 4

Purpose

.....

.....

[8]

9626/12 Feb/March 2025 · Question 12

[Mark scheme](#)

12 Employees log in to their company's database with a User ID and password.

A login screen is displayed. A user enters their User ID.

If employees have forgotten their password, they click on a 'Forgot Password?' button.

Draw a program flowchart to represent the following process.

- If the 'Forgot Password?' button is clicked, a message box appears and asks them to enter their User ID.
- A check will be performed to see if the system recognises the User ID.
- If the User ID is **not** recognised, an error message is displayed and the user will be returned to the start of the process.
- If the User ID is recognised, a security question is asked.
- The user's response to the security question will be checked. If the user's response is correct, then an email containing the password reminder will be sent to the user's email address.
- If an incorrect response is given, an error message will be displayed and the user will be returned to the start of the process.

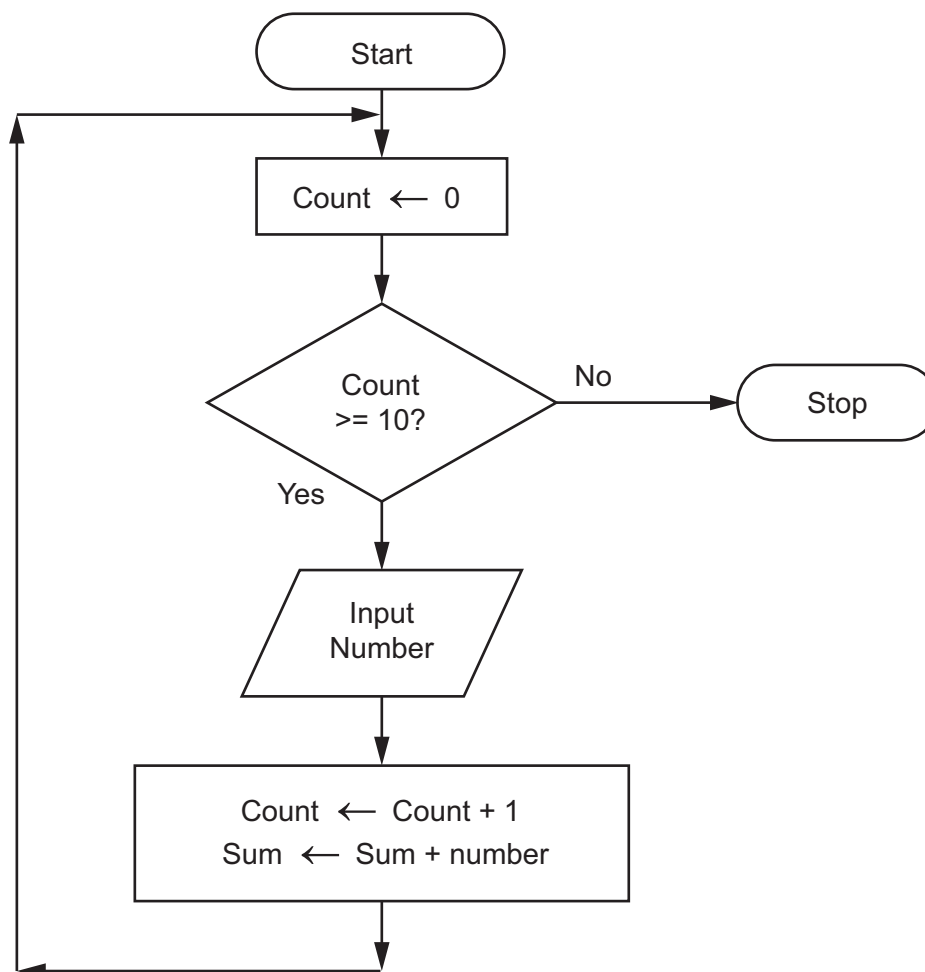
Draw your program flowchart on the opposite page.

[8]

Draw your program flowchart below.

- 9 The program flowchart produces the output of the sum of 10 numbers.

There are errors and omissions in the flowchart.



Explain **two** corrections that should be made to the flowchart.

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..... [4]

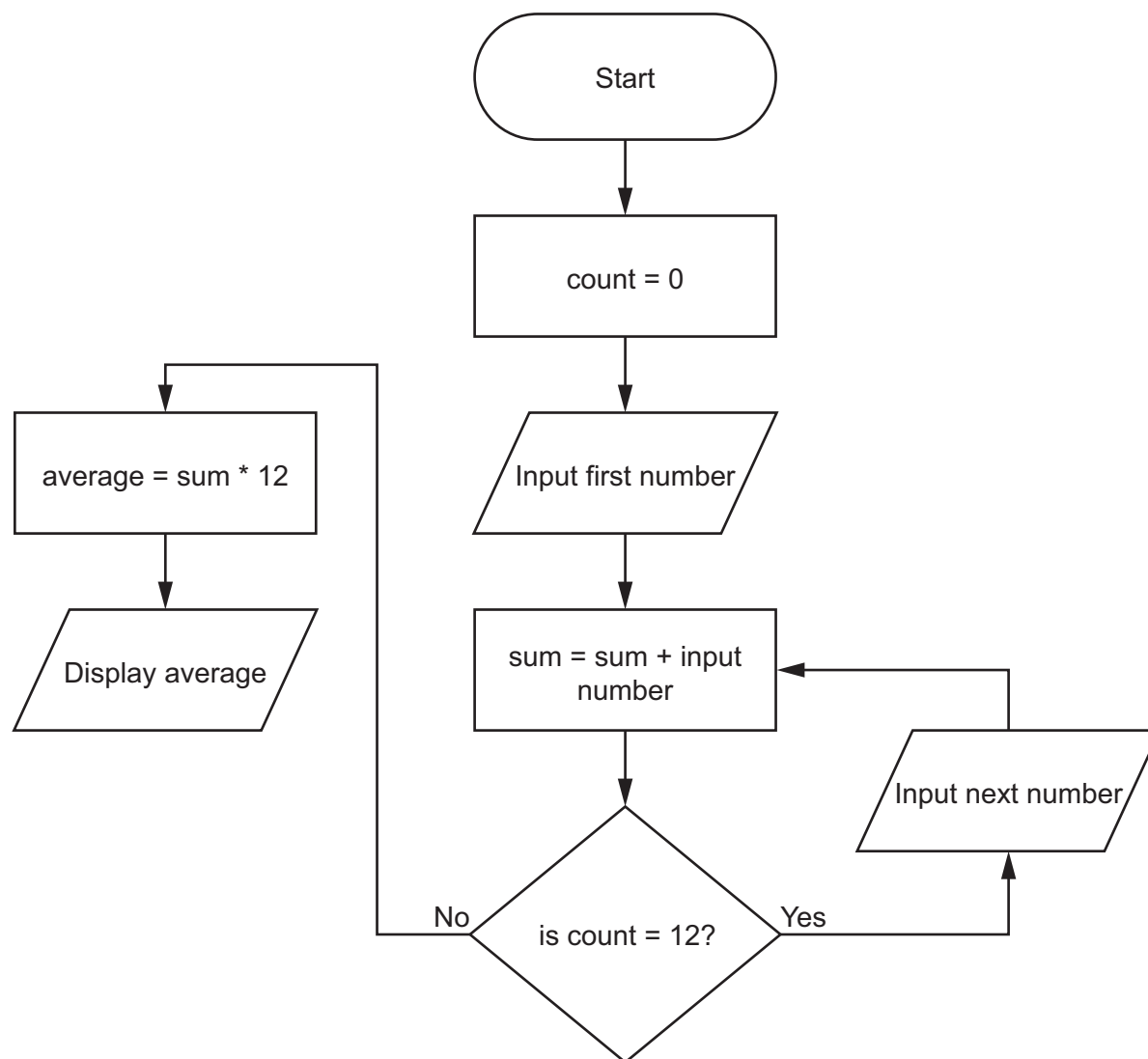
Write a pseudocode algorithm using REPEAT...UNTIL which allows the teacher to input 20 test marks and outputs the correct message for each inputted mark.

[8]

13 The average (mean) of a set of 12 numbers is calculated by:

- adding the numbers together to get a total
- dividing the total by 12.

A student has attempted to draw a flowchart of an algorithm which finds the average of 12 numbers. They have made some errors.



Redraw the flowchart, with the errors corrected.

[5]

- 5 Lucia has written an algorithm in pseudocode to determine whether a number (X) is a factor of another number (Y).

X is a factor of Y if the result of Y/X is a whole number.

- The algorithm uses a WHILE...ENDWHILE statement to check whether X is greater than zero and that X is less than Y.
 - If X is greater than zero and X is less than Y, the algorithm will output:
 - the values of X and Y
 - an appropriate message if X is a factor of Y
- OR
- an appropriate message if X is **not** a factor of Y.

In pseudocode `INT (W)` returns the whole number part of W.

```

INPUT X
INPUT Y
WHILE X=0 AND <=Y
W←Y/X
Z=INT(W)
IF W=Z
    THEN
        PRINT X, " is a factor of Y"
    ELSE
        PRINT X, " is not a factor of ",Y
ENDWHILE

```

Lucia has made **four** mistakes in her algorithm.

Identify **four** mistakes in the code. For each mistake, explain in detail what the code as written would achieve.

Stating that the code would not work is not sufficient to gain any marks.

Mistake 1

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Mistake 2

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Mistake 3

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.....

Mistake 4

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[8]

7 A student has created an algorithm to calculate the average of a set of numbers.

- The user will be asked how many numbers they wish to enter
- The user will respond with a number
- The user will then enter that many numbers
- The algorithm will total the numbers entered and divide the total by how many numbers the user said they will enter.

The algorithm is shown. The line numbers have been included to help you.

```

10 total ← 10
20 PRINT "Input number of repeats"
30 INPUT n
40 count → 1
50 REPEAT
60     count ← + 1
70     PRINT "Input number ", count
80     INPUT number
90     total + number ← total
100 UNTIL count = number
110 average ← total / n
120 PRINT average

```

The student has made some mistakes in the algorithm.

(a) State **three** mistakes in this algorithm and suggest how each can be corrected.

mistake 1

.....

.....

.....

mistake 2

.....

.....

.....

mistake 3

.....

.....

.....

[6]

This algorithm uses a REPEAT...UNTIL loop. The student wishes to change the algorithm so that it uses a WHILE...ENDWHILE loop.

(b) Contrast a REPEAT...UNTIL loop with a WHILE...ENDWHILE loop.

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..... [2]

9626/11 May/June 2026 · Question 8

[Mark scheme](#)

- 8** A combined heating and air conditioning system has been installed in a house. The system measures the temperature in the house and reacts as programmed. The user can override the temperature settings of the system to suit their needs.

(a) Identify **two** input devices that will be used as part of this system.

input device 1

input device 2 [2]

(b) Explain, with the example of a pump, the purpose of an actuator in this system.

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..... [2]

The system has the following rules:

- If the temperature exceeds 28 degrees Celsius, a fan will turn on and the heating system will turn off.
 - If the temperature falls below 19 degrees Celsius, the fan will turn off and the heating will turn on.
- (c) Complete the pseudocode algorithm to show the processing that takes place in this combined system. The variable `temperature` represents the temperature of the house.

```
REPEAT  
    INPUT temperature
```

- 2 A partially completed algorithm is shown. Line numbers are given for reference. Some lines are incomplete.

The user will input five numbers, one number at a time.

After five numbers have been entered, the largest number and the smallest number will be displayed as part of a suitable message.

The algorithm uses a WHILE loop.

```

10  Count  ←    0
20  Number ←    0
30  Largest_Number ← 0
40  Smallest_Number ← .....
50  INPUT Number
60  Largest_Number ← Number
70  Smallest_Number ← Number
80  WHILE Count < .....
90      INPUT .....
100     IF Number ..... Largest_Number
110         .....
120             Largest_Number ← Number
130         .....
140             IF Number ..... Smallest_Number
150                 THEN Smallest_Number ← Number
160             ENDIF
170         .....
180     Count  ←    .....
190 ENDWHILE
200 PRINT "The largest number is "Largest_Number"
      and the smallest number is "Smallest_Number"."

```

- (a)** Use the table to show how each line may be completed. Line 40 has been done for you as an example of how the question should be answered.

line	current code	completed code
40	Smallest_Number $\leftarrow$	Smallest_Number $\leftarrow$ 0
80	WHILE Count <	
90	INPUT	
100	IF Number Largest_Number	
110		
130		
140	IF Number Smallest_Number	
170		
180	Count $\leftarrow$	

[8]

- (b)** Explain why `Largest_Number` (line 60) and `Smallest_Number` (line 70) are both set to the same value collected in line 50.

..... [4]

[4]

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 7

[Back to question](#)

Question	Answer	Marks
7	Eight from: - Input number of numbers - Initialise count - Initialise total - Loop that works involving REPEAT... UNTIL (e.g. count = n) - Input number within loop - Increment count within loop - Update total within loop - Calculation of average - output average Possible solution could be: <pre> INPUT n count ← 0 total ← 0 REPEAT INPUT number count ← count + 1 total ← total + number UNTIL count = n average ← total / n PRINT average </pre>	8

Question	Answer	Marks
6	Six from: <pre> REPEAT INPUT temperature IF temperature < preset THEN IF pump switched off THEN send signal to switch pump on ENDIF ELSE IF pump switched on THEN send signal to switch pump off ENDIF ENDIF UNTIL system switched off </pre> 1 mark for correct 1st IF statement 1 mark for correctly positioned first action 1 mark for correctly positioned 1st ENDIF 1 mark for correct 2nd IF statement 1 mark for correctly positioned second action 1 mark for correctly positioned 2nd ENDIF 1 mark for correctly positioned 3rd ENDIF 1 mark for correct UNTIL statement... ...1 for correct position of correct UNTIL statement	6

Question	Answer	Marks
7	<pre> graph TD Start([Start]) --> Init[count ← 0 max ← 0] Init --> Input[/Input number/] Input --> Dec1{Is number ≤ max?} Dec1 -- yes --> IncCount[count ← count + 1] Dec1 -- no --> UpdateMax[max ← number] UpdateMax --> IncCount IncCount --> Dec2{Is count = 10?} Dec2 -- yes --> Print[/Print max/] Dec2 -- no --> Input Print --> Stop([Stop]) </pre> Is number $\leq$ max in 1st decision box – 1 mark Yes/no in correct positions at 1st decision box – 1 mark max $\leftarrow$ number in correct box – 1 mark increment count in correct box – 1 mark is count = 10? in 2nd decision box – 1 mark yes/no in correct positions at 2nd decision box – 1 mark Print max in correct box – 1 mark	7

Question	Answer	Marks
9(a)	<pre> READ first record in transaction file READ first record in old master file WHILE not end of transaction file IF transaction file WorkerID = master file WorkerID THEN WagesSoFar ← WagesSoFar + (RateOfPay*HoursWorked) WRITE updated master file record to new master file READ next record in transaction file ENDIF READ next record in master file ENDWHILE </pre> Six from: Correct calculation – RateOfPay*HoursWorked – 1 mark Correct calculation - WagesSoFar ← WagesSoFar + (RateOfPay*HoursWorked) – 1 mark Correct update – 1 mark Read next transaction file record... – 1 mark ...immediately before ENDIF – 1 mark ENDIF– 1 mark Read next master file record... – 1 mark ... immediately after ENDIF – 1 mark ENDWHILE at end of algorithm – 1 mark	6

Question	Answer	Marks
9(b)	Eight from: <i>Advantages</i> Batch processing does not require constant use of the computer when it might be needed for other tasks... ...the payroll only needs to be run at the end of the week/month... ...whereas real time processing would require computer to be available all the time... ...batch processing/payroll can be run at convenient times for the company/when the computing resources are less busy/when there is little demand for computer resources/can be run overnight Real-time processing requires immediate processing of data which is unnecessary with payroll... ...as hours worked would only be known at the end of each day so no point in using real-time processing Real-time processing is not usually capable of processing large volumes of data such as running a payroll whereas batch processing is Batch processing systems only need lower specification computers... ...whereas real-time processing needs more complex computer systems Online processing requires more hardware making it more expensive than batch processing Batch processing only needs a few employees to enter the data... ...so total wage bill is less... ...whereas online processing systems can need employees to be constantly using the system In batch processing systems there are fewer/no transcription/update/data errors... ...than human operators would produce in an online system <i>Disadvantages</i> With batch processing there is a time delay between collecting the input data and getting an output/data is not processed until the time specified With batch processing the information is only up to date after the master file has been updated by the transaction file... ...whereas in real-time processing information is always up to date With batch processing systems errors are only revealed during processing/after batch process is complete which can be overnight and so cause delays/errors cannot be corrected until the batch process is complete... ...in online processing errors are revealed, and can be acted upon, immediately Users cannot terminate a process during execution/have to wait until execution is completed Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Question	Answer	Marks
8	REPEAT <i>INPUT reading from induction loop</i> IF reading = pre-set value THEN send signal to raise barrier REPEAT IF light sensor activated THEN send signal to lower barrier ENDIF UNTIL light sensor activated ENDIF UNTIL system switched off Seven from: Appropriate IF statement (comparison of reading) Send signal to raise barrier after first IF Appropriate other IF statement (checking if light sensor activated) Send signal to lower barrier after 2nd IF Appropriate position and matching 1st ENDIF Appropriate position and matching 2nd ENDIF Appropriate UNTIL system switched off statement Appropriate loop checking that light sensor has been activated	7

Question	Answer	Marks
11	<pre> graph TD Start([Start]) --> Init[count ← 0 total ← 0] Init --> Input[/INPUT number/] Input --> Loop[count ← count + 1 total ← total + number] Loop --> Decision{is count = 6?} Decision -- no --> Input Decision -- yes --> Print[/PRINT total/] Print --> Stop([Stop]) </pre> Initialise count – 1 mark Initialise total – 1 mark total $\leftarrow$ total + number – 1 mark count $\leftarrow$ count + 1 – 1 mark is count = 6 – 1 mark PRINT total – 1 mark yes and no in correct positions – 1 mark	7

Question	Answer	Marks
9(a)	One from: - Electrical resistance is changed/voltage is generated (depending on the temperature) (1) Electrical signals are generated (which are converted into values (to represent temperature)) (1).	1
9(b)	<pre> WHILE system switched on INPUT temperature IF temperature > preset THEN IF window closed THEN send signal to motor to open window ENDIF ELSE IF window open THEN send signal to motor to close window ENDIF ENDIF ENDIF ENDWHILE </pre> - send signal to open//open window (NOT 'output') (1) - send signal to open//open window as 1st statement (1) - ENDIF closes the IF statement (1) - ALL three ENDIFs in same vertical plane as relevant IFs (within reason) (1) - ELSE (1) - ELSE in same vertical plane as THEN (1) 'IF window open' (or equivalent) ... (1) ... IF window open condition closed with ENDIF (1) - Close window instruction (1) - Send signal to motor to close window appropriately positioned (1) - IF temperature > preset value condition closed with ENDIF (1) - Whole of IF Window open section indented correctly (allow follow through on close window instruction position) (1) - ENDWHILE as final statement (1)	8

Question	Answer	Marks
9(c)	Eight from: Advantages max 6 • Reduced chance of plants dying/wilting etc (1) • Jose could forget to (e.g.) open window//be so busy that he is unable to (e.g.) open window ... (1) ... computers can be permanently switched on and take action regularly/will not forget (1) • Jose can do other tasks (1) • Computers can take action immediately if the temperature gets too hot ... (1) ... there would be a delay before Jose can take action (1) • Readings taken by computers tend to be more accurate ... (1) ... and action taken is therefore more likely to be the correct one (1) (ACCEPT CONVERSE ARGUMENTS ABOUT JOSE'S INACCURACY) • If Jose was physically disabled, he may find it difficult to open window ... (1) ... whereas computer can do it automatically (1) Disadvantages max 6 • Computers can be expensive to buy ... (1) ... whereas it would not cost Jose any money to look after the plants himself (1) • Computer maintenance costs can be expensive ... (1) ... whereas with Jose that is not a problem (1) • It is quite difficult to program computers to take correct action ... (1) ... but Jose would know instinctively//will have expertise (1) • Sensors can deteriorate//become inaccurate ... (1) ... whereas Jose's actions will tend to be more consistent (1) ... and will need recalibration (1) • A computer-controlled system will not be able to function if there is a problem with the computer/there is a power cut/load shedding (1) Max. six marks if bullets/list of points Must have at least expansions or comparisons to achieve six marks in either section	8

Question	Answer	Marks
9(a)	<pre>PROCEDURE AfterTax(WagesBeforeTax) TaxPaid ← WagesBeforeTax * 35% WagesAfterTax ← WagesBeforeTax - TaxPaid ENDPROCEDURE</pre> • PROCEDURE and Appropriate name as first statement (ignore any gaps) (1) • WagesBeforeTax in brackets on first line (1) • Calculations and assignments within the procedure result in the correct numerical outcome for the question (Ignore errors with variable names, as long as it is clear which variable is being referred to) (1) • Sensible variable names with no gaps (WagesBeforeTax has been given, must be at least TWO extra variables) (1) • ENDPROCEDURE as last statement (1)	4

Question	Answer	Marks
9(b)	<pre>count ← 0 INPUT NumberOfWorkers REPEAT</pre> • INPUT Hours, Rate (1) (Candidate may do this in two steps) • INPUT Hours, Rate as first line/before BeforeTax is called (allow follow through) (1) • (CALL) BeforeTax(Hours, Rate) (1) • (CALL) AfterTax(WagesBeforeTax) (1) • BeforeTax before WagesAfterTax (1) • PRINT/OUTPUT WagesAfterTax (1) (Must be in correct place) • count ← count + 1 (1) (can be anywhere) • UNTIL count = NumberOfWorkers (1)	6

Question	Answer	Marks
8	Six marks available: - Any indication that an ENDIF is missing/needs to be added (1) Line 6 should be $\geq$ (1) Line 8 B should be in quotes//it should be 'B' (1) - This only allows one student's marks to be entered (1) - There should be an INPUT of the number of students ... (1) ... as the 1st line/before INPUT mark (1) - Any indication that a COUNT is required ... (1) ... Count should be initialised at the second line/3rd line (1) - Any indication that iteration is required ... (1) ... A loop should be set up as the third statement/ after the 1st input and count has been initialised (1) - The count should be incremented within the loop (1) - The loop should be closed as the last statement (1) - Numbers between 50 and 60 would not have a grade//no B grades would be awarded (1) Where candidate has not identified the location of the error/improvement required they can include direct quotes from the code to make it clear to which section they are referring. Where candidate has done so, if their answer is the corrected version of the code, award the mark.	6

Question	Answer	Marks
4(a)	<pre> graph TD Start([Start]) --> D1{Is the system activated?} D1 -- no --> Stop([Stop]) D1 -- Yes --> I1[/Input mlevel/] I1 --> D2{is mlevel > dry?} D2 -- no --> D3{is sprinkler off? R} D2 -- yes --> D4{is sprinkler on?} D4 -- yes --> O1[/send signal to actuator to switch sprinkler off/] D4 -- no --> D3 O1 --> J1(()) D3 -- yes --> O2[/send signal to actuator to switch sprinkler on/] D3 -- no --> J1 O2 --> J1 J1 --> D1 </pre> SIX marks available: • Yes/No in correct position off first decision box (1) • Two other Yes/No used (off a decision) (1) – A further Yes/No used (off a decision) (1) • <u>Input MLevel</u> in correct place (1) • <u>Is sprinkler on/is sprinkler off</u> added in correct place (1) MLevel>dry NO branch: • Is Sprinkler OFF – YES – switch <u>Sprinkler ON</u> in correct place (1) OR • Is Sprinkler ON – NO – switch <u>Sprinkler ON</u> in correct place (1) MLevel >dry YES branch • Is Sprinkler ON -YES- switch <u>Sprinkler OFF</u> in correct place (1) OR • Is Sprinkler OFF – NO – switch <u>Sprinkler OFF</u> in correct place (1) For final four mark points, marks may only be awarded where the logic leading up to the decision matches that given here and is shown in full.	

Question	Answer	Marks			
4(b)(i)	<table border="1" style="width: 100%; height: 80px;"><tr><td style="width: 15%;"></td><td style="width: 70%;"></td><td style="width: 15%;"></td></tr></table>				1
4(b)(ii)	ONE from: - The name of the subroutine (1) - Start (1)	1			

Mark scheme · 9626/11 May/June 2023 · Question 8

[Back to question](#)

Question	Answer	Marks
8	<pre> graph TD Start([Start]) --> Init[OT ← 0 ET ← 0 count ← 0] Init --> LoopStart(()) LoopStart --> CountInc[count ← count + 1] CountInc --> InputN[/Input N/] InputN --> XCalc[X ← N / 2] XCalc --> IsInt{Is/If X = INT(X)?} IsInt -- No --> OTAdd[OT ← OT + N] IsInt -- Yes --> ETAdd[ET ← ET + N] OTAdd --> Count10{count = 10?} ETAdd --> Count10 Count10 -- No --> LoopStart Count10 -- Yes --> Print[/Print OT, ET/] Print --> Stop([Stop]) </pre>	8

Mark scheme · 9626/12 May/June 2023 · Question 10

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Question	Answer	Marks
10	Six marks available <pre> count ← 0 smallest ← 1000 WHILE count < 6 INPUT number IF number < smallest THEN smallest ← number ENDIF count ← count + 1 ENDWHILE PRINT smallest </pre>	6

Mark scheme · 9626/13 May/June 2023 · Question 11

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Question	Answer	Marks
11	<pre> graph TD Start([Start]) --> Input[/Input earnings/] Input --> D1{earnings >= 10 000} D1 -- no --> Tax0[tax ← 0] D1 -- yes --> D2{earnings > 40 000} D2 -- no --> Tax20[tax ← (earnings - 10 000) * 20%] D2 -- yes --> Tax40[tax ← 6000 + ((earnings - 40 000) * 40%)] Tax0 --> Net[net ← earnings - tax] Tax20 --> Net Tax40 --> Net Net --> Print[/Print net/] Print --> Stop([Stop]) </pre>	8

Question	Answer	Marks
7	Eight from: 1 mark for each correct statement <pre> count ← 0 INPUT number REPEAT INPUT jobcode, hours CASE (1 - present and location) OF jobcode (1 - allow on FT) 'L' : rate ← 18 'O' : rate ← 22 'M' : rate ← 30 OTHERWISE (1): PRINT/OUTPUT "Invalid job code" ENDCASE CASE and ENDCASE correctly indented wage ← hours * rate statement outside of CASE statement PRINT/OUTPUT wage (present) PRINT/OUTPUT wage (outside of CASE statement) count ← count + 1 UNTIL count = number UNTIL count = number as final statement </pre>	8

Question	Answer	Marks
9	Max Eight: <pre> count ← 0 smallest ← 1000 largest ← 0 WHILE count < 10 INPUT base, height CALL (must NOT have arrow, or colon) (1) parallelogram (base, height) (may award as FT if error in CALL or in wrong place) IF area < smallest THEN smallest ← area ELSE (appears here even if the THEN statement is not completed/is wrong) IF area > largest THEN (appears here (IF test may be wrong)) largest ← area ENDIF IF and ENDIF indented correctly ENDIF count ← count + 1 (if present) count ← count + 1 (if within while statement but not if within IF statement (unless in both) (allow FT for structure from previous mark point)) ENDWHILE PRINT/OUTPUT smallest (1) largest (Do not accept OUTPUT smallest and largest) ANY PRINT/OUTPUT statement as final statement </pre>	8

Mark scheme · 9626/13 Oct/Nov 2023 · Question 8

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Question	Answer	Marks
8	Six from: - Immediately before/after/inside $\text{count} \leftarrow 0$ box insert $\text{total} \leftarrow 0$ (1) - Immediately before/after/inside $\text{total} \leftarrow \text{total} + \text{number}$ box insert $\text{count} \leftarrow \text{count} + 1$ (1) $\text{average} \leftarrow \text{total}/\text{number}$ should be changed to $\text{average} \leftarrow \text{total}/10$ (1) OR $\text{average} \leftarrow \text{total}/\text{count}$ (1) - yes/no should be reversed (1) OR $\text{count}=10$ changed to $\text{count}<10$ (1) - Loop line should return to flowchart above Input number box (where drawn on the flowchart, no need to delete the original line or add arrow heads) (1) - Print/output average should be included after $\text{average} \leftarrow \text{total}/10$ box (1) - Stop box/end should be included at end of flowchart (1) - Add instructions to input box (1)	6

Mark scheme · 9626/13 Oct/Nov 2023 · Question 9

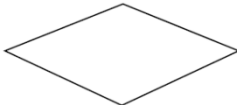
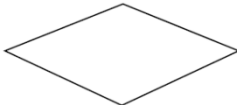
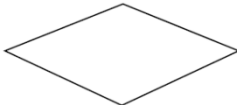
[Back to question](#)

Question	Answer	Marks
9	<pre> INPUT X, Y W←X/Y Z=INT(W) IF W=Z THEN PRINT/OUTPUT Y, " (e.g.) is a factor of ", X ELSE PRINT/OUTPUT Y, " (e.g.) is not a factor of ", X ENDIF </pre> Eight from: - INPUT X (1) Y (1) - Correct IF statement (1) - IF and ENDIF indented correctly (1) - Use of THEN (1) - Valid output statement (1) with both commas (1) with speech marks (1) - Use of ELSE (1) - THEN and ELSE indented correctly (1) - Valid output statement after ELSE (1) Commas between values and text in second statement text within speech marks (1) - Correct use of ENDIF (1) - No further code after ENDIF statement (1)	8

Question	Answer	Marks
8	Six from: e.g.: INPUT number CASE number >12 : X ← number / 12 >8 : X ← number / 8 OTHERWISE: X ← number / 4 ENDCASE OUTPUT X • INPUT number (name of variable is not important) (1) • INPUT number as first item (1) • CASE number (any variable, but must match MP1 above) in correct place (1) • Action if >12 correct plus calculated output (1) • Action >=8<=12 correct (1st) plus calculated output (MUST be >= or <= structure) (1) • Action <4 correct plus calculated output (1) • ENDCASE in correct position (1) • CASE and ENDCASE correct indentation (1) • ALL other correct aspects indented from CASE and ENDCASE (1)	6

Question	Answer	Marks
7	Eight from: Example of program flowchart <pre> graph TD Start([Start]) --> Input[/INPUT Mark/] Input --> D1{Mark < 85?} D1 -- No --> P1[/PRINT "Grade A"/] D1 -- Yes --> D2{Mark < 70?} D2 -- No --> P2[/PRINT "Grade B"/] D2 -- Yes --> D3{Mark < 50?} D3 -- No --> P3[/PRINT "Grade C"/] D3 -- Yes --> P4[/PRINT "Grade U"/] P1 --> Stop([Stop]) P2 --> Stop P3 --> Stop P4 --> Stop </pre> 1 START and STOP terminators 1 mark 2 Input Mark 1 mark 3 A, B and C grade decisions logically correct (check order) 2 marks OR 4 Two of A,B or C decisions logically correct (check order) 1 mark 5 All decision box used have Yes and No 1 mark 5 A, B and C grades output from relevant logic 1 mark 6 U Grade output from correct logic AND correct yes/no decision 1 mark 7 All flow lines correctly located 1 mark 8 All lines have arrow heads with direction of flow 1 mark 9 All correct symbols/shapes 1 mark	8

Question	Answer	Marks
4	Example solution: <pre>graph TD Start([START]) --> Input[/INPUT Num/] Input --> SetMax[SET MAX to Num] SetMax --> SetTotal[Set Total to Num] SetTotal --> SetCount[Set Count to 1] SetCount --> IsCount{Is Count = 21?} IsCount -- YES --> SetAvg[Set Avg to Total/20] SetAvg --> Output[/Output MAX, Avg/] Output --> End([END]) IsCount -- NO --> Input2[/INPUT Num/] Input2 --> SetTotal2[Set Total to Total + Num] SetTotal2 --> IsNumMax{Is Num > MAX?} IsNumMax -- YES --> SetMax2[Set MAX to Num] IsNumMax -- NO --> CountInc[Count ← Count + 1] SetMax2 --> CountInc CountInc --> IsCount</pre> SEVEN marks available: 1 Start and End (accept STOP) terminators (1) 2 TWO Correctly drawn decision boxes with YES/NO decisions (1) 3 All YES/NO decisions (used) go to correct outcomes (i.e. all logic is correct) (1) 4 Input box and output box are both parallelograms (1) 5 Process box(es) are rectangles (1) 6 Two variables (e.g. MAX, COUNT, TOTAL, NUM) initialised/set to zero before loop (1) 7 Input value from user (attempt at) (1) 8 Total set to first input value (1) 9 Inputs limited to 20 values (attempt at use of logic to find) (1) 10 Sum all values input (1) 11 Set new MAX when input value > MAX in loop 12 Increment COUNT (1) 13 Correctly calculate average (must include process/calculation) (1) 14 Output MAX and average (1) (This may happen anywhere, but if in loop, must be after any new number is added).	7

Question	Answer	Marks																
1	EIGHT from: 1 mark for each correct symbol (1st mark) (max 4) 1 mark for each correct matching purpose stated (max 4) Taken from syllabus: <table><tr><th><u>Purpose</u></th><th><u>Symbol</u></th></tr><tr><td>Input/output</td><td></td></tr><tr><td>Decision</td><td></td></tr><tr><td>Terminator (Start/Stop)</td><td></td></tr><tr><td>Process box</td><td></td></tr><tr><td>Subroutine</td><td></td></tr><tr><td>Connector</td><td></td></tr><tr><td>Flow line</td><td></td></tr></table>	<u>Purpose</u>	<u>Symbol</u>	Input/output		Decision		Terminator (Start/Stop)		Process box		Subroutine		Connector		Flow line		8
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Input/output																		
Decision																		
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Process box																		
Subroutine																		
Connector																		
Flow line																		

Question	Answer	Marks
12	One mark per bullet point to a maximum of eight marks. - Start and stop correct shape (1) - Two relevant questions asked in diamond shape (1) - Third relevant question asked in diamond shape (1) - Any use of YES and NO leading from any relevant question (1) - Any example of Yes and No logic BOTH correct (logic must match the question being asked) (1) - User ID input message (1) - Password reminder input message (1) - User ID Error message (1) - Answer Error message (1) - Email/message sent (1) - Loop back to start (from any error) (1) - Inputs and outputs in parallelograms (may be 1) (1)	8

Question	Answer		Marks
9	One mark per bullet point to a maximum of four marks.		4
	Error (MAX 2)	Answer – MUST identify what is wrong with the code OR what the new code now does, and justify the change	
	Count is incorrect for the logic – this error identified (1st)	e.g. - Count > 10 means that the code ends on first run through AND - Change to Count < 10 OR - Change to Count = 9 (1) - Candidate may answer by arguing that Count should have been set to 1. This is acceptable, but this expansion may only be marked once.	
	SUM variable not initialised (1st)	- SUM ← 0 added to process box AND - SUM + calculation will not work, as no initial value for SUM (1)	
	No output (1st)	- The flowchart will not output a value for the sum of the numbers - AND - OUTPUT added for SUM along NO path (1)	
	Loop goes to wrong place (1st)	- Flowchart will run without end OR - Loop goes to wrong part of the flowchart AND - Loop changed to go after initialisation of COUNT/before decision box (1)	

Question	Answer	Marks
7	<i>Example solution:</i> <pre> Count ← 0 REPEAT INPUT mark IF mark > 70 THEN OUTPUT "You have achieved a Distinction" ENDIF IF mark >= 55 AND <= 70 THEN OUTPUT "You have achieved a Merit" ENDIF IF mark >= 40 AND <= 54 THEN OUTPUT "You have achieved a Pass" ENDIF IF mark < 40 THEN OUTPUT "You have not reached the Pass mark" ENDIF Count ← Count + 1 UNTIL Count = 20 </pre> Eight marks from: • Initialise Count before Mark variable is input (1) • REPEAT in correct logical location (1) • Input Mark (1) • Correct Distinction grade criterion//correct fail criterion (1) • Correct Merit grade criterion//correct Pass grade criterion (1) • Correct Increment of Count (1) • Increment of Count is within the loop (1) • UNTIL Count = 20 (1) • UNTIL in correct position (1) • Use of one IF...ENDIF (1) • All OUTPUTs correctly match the criteria (NOT awardable if BP4 or BP5 not awarded) (1)	8

Question	Answer	Marks
13	Max five marks from: • Add sum = 0 (sum initialisation) (1) • Add count = count + 1 (iteration of count) (1) • Replace average calculation from * to / (1) • Reverse Yes/No correctly (1) • Add end shape (1) Corrected flowchart: <pre> graph TD Start([Start]) --> Sum0[sum = 0] Sum0 --> Count0[count = 0] Count0 --> Input1[/Input first number/] Input1 --> SumAdd[sum = sum + input number] SumAdd --> CountInc[count = count + 1] CountInc --> Decision{is count = 12?} Decision -- Yes --> AvgCalc[average = sum / 12] AvgCalc --> Display[/Display average/] Display --> End([End]) Decision -- No --> InputNext[/Input next number/] InputNext --> SumAdd </pre>	5

Question	Answer	Marks														
5	Eight available. Two per error and improvement. ONLY consider a maximum of four errors. <table><tr><td>Location (1st mark)</td><td>What would be achieved</td></tr><tr><td>WHILE X=0</td><td>WHILE loop would continue if X is equal to zero/would not continue if x is greater than 0</td></tr><tr><td>AND <=Y</td><td>Logic error as <Y is not being compared to anything</td></tr><tr><td><=Y</td><td>If X = Y, algorithm will continue.</td></tr><tr><td>PRINT X, "is a factor of Y"</td><td>Would not show value of Y / would show Y as the value</td></tr><tr><td>Lines of code following the WHILE statement are NOT indented</td><td>Code would run (regardless if conditions are met) ONCE/no loop</td></tr><tr><td>No ENDIF</td><td>The IF statement will only consider one line/will not compute as no close detected</td></tr></table>	Location (1st mark)	What would be achieved	WHILE X=0	WHILE loop would continue if X is equal to zero/would not continue if x is greater than 0	AND <=Y	Logic error as <Y is not being compared to anything	<=Y	If X = Y, algorithm will continue.	PRINT X, "is a factor of Y"	Would not show value of Y / would show Y as the value	Lines of code following the WHILE statement are NOT indented	Code would run (regardless if conditions are met) ONCE/no loop	No ENDIF	The IF statement will only consider one line/will not compute as no close detected	8
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PRINT X, "is a factor of Y"	Would not show value of Y / would show Y as the value															
Lines of code following the WHILE statement are NOT indented	Code would run (regardless if conditions are met) ONCE/no loop															
No ENDIF	The IF statement will only consider one line/will not compute as no close detected															

Question	Answer	Marks
7(a)	SIX marks available, TWO per identified and corrected error: <pre> 10 total ← 10 (1) (Incorrect initial value//value should be 0//value does not need to be set) (1) 20 PRINT "Input number of repeats" 30 INPUT n 40 count → 1 (1) (Count arrow pointing in wrong direction) (1) (Count value should be 0 (this mark may be claimed at line 100)) (1) 50 REPEAT 60 count ← + 1 (1) (Incorrect statement for increment) (1) 70 PRINT "Input number ",count 80 INPUT number 90 total + number ← total (1) (Incorrect order//setting wrong value to variable) (1) 100 UNTIL count = number (1) (Incorrect statement for comparison/should compare to n) (1) 110 average ← total / n 120 PRINT average </pre>	6

Question	Answer	Marks
7(b)	TWO marks available: • A WHILE...ENDWHILE loop will only be executed if the condition is true (1) BUT a REPEAT...UNTIL loop will happen even it is not true. (1) • A REPEAT...UNTIL loop is guaranteed to happen, (1) but a WHILE...ENDWHILE loop may not happen. (1) • For a WHILE...ENDWHILE loop, the condition is tested before the statement runs (1) for a REPEAT...UNTIL loop, the condition is tested after the statement runs. (1) • A WHILE...ENDWHILE loop is a pre-condition loop, (1) and a REPEAT...UNTIL loop is a post-condition loop. (1) • A REPEAT ...UNTIL loop ends when a condition is met (1) but a WHILE...ENDWHILE loop ends when a condition is not met. (1)	2

Question	Answer	Marks	Guidance
8(a)	TWO marks available: • Temperature sensor/Thermostat (1) • Keypad (1) • Microphone (1) • Touch screen (1) • (Push) Switch (1) • Remote control (1)	2	Do Not Accept: • Smart Phone etc. • Button • Heat sensor
8(b)	TWO marks available: • The actuator converts a signal <u>from the microprocessor</u> into an action/to perform an action (1) • To turn the pump on/off (1)	2	

Question	Answer	Marks	Guidance																						
8(c)	EIGHT marks available: <table><tr><td>IF temperature >28</td><td>1</td></tr><tr><td>THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)</td><td>1</td></tr><tr><td>(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u></td><td>1</td></tr><tr><td>Send signal to turn on indented</td><td>1</td></tr><tr><td>ENDIF correctly positioned within the algorithm</td><td>1</td></tr><tr><td>IF temperature <19 with matching ENDIF</td><td>1</td></tr><tr><td>THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u></td><td>1</td></tr><tr><td>Any use of ENDIF when in capitals and not split</td><td>1</td></tr><tr><td>UNTIL SYSTEM SWITCHED OFF</td><td>1</td></tr><tr><td>UNTIL statement correctly positioned and indented</td><td>1</td></tr><tr><td>ALL IFs and ENDIFs correctly indented</td><td>1</td></tr></table>	IF temperature >28	1	THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)	1	(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u>	1	Send signal to turn on indented	1	ENDIF correctly positioned within the algorithm	1	IF temperature <19 with matching ENDIF	1	THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u>	1	Any use of ENDIF when in capitals and not split	1	UNTIL SYSTEM SWITCHED OFF	1	UNTIL statement correctly positioned and indented	1	ALL IFs and ENDIFs correctly indented	1	8	
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Question	Answer	Marks	Guidance																											
2(a)	EIGHT marks available: <table><tr><th>Line</th><th>Item to be added</th><th></th></tr><tr><td>80</td><td>4</td><td>1</td></tr><tr><td>90</td><td>Number (accept lower case)</td><td>1</td></tr><tr><td>100</td><td>></td><td>1</td></tr><tr><td>110</td><td>THEN (accept lower case)</td><td>1</td></tr><tr><td>130</td><td>ELSE (accept lower case). Do not award ELSEIF</td><td>1</td></tr><tr><td>140</td><td><</td><td>1</td></tr><tr><td>170</td><td>ENDIF (DO NOT accept lower case or if split)</td><td>1</td></tr><tr><td>180</td><td>Count + 1 (accept lower case)</td><td>1</td></tr></table>	Line	Item to be added		80	4	1	90	Number (accept lower case)	1	100	>	1	110	THEN (accept lower case)	1	130	ELSE (accept lower case). Do not award ELSEIF	1	140	<	1	170	ENDIF (DO NOT accept lower case or if split)	1	180	Count + 1 (accept lower case)	1	8	Each answer is CAO ONLY considers answers on the table
Line	Item to be added																													
80	4	1																												
90	Number (accept lower case)	1																												
100	>	1																												
110	THEN (accept lower case)	1																												
130	ELSE (accept lower case). Do not award ELSEIF	1																												
140	<	1																												
170	ENDIF (DO NOT accept lower case or if split)	1																												
180	Count + 1 (accept lower case)	1																												
2(b)	FOUR from: - It is the only number that has been entered at that stage (1st) there is no comparison to be made (1) (award once only) - It must be both largest and smallest (1st) there is no comparison to be made//any comparison would be comparing a number with itself (1) (award once only) - Ensures that the highest/lowest mark recorded is valid (1) - Overwriting both set values with a value inputted by the user (1) - One mark for a numerical example of either lower or higher number being ignored (1)	4																												